

MATH115A Written Exercises

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1 Q

We want to show that $0 \cdot v = 0$.

Notice that

$$0v + v = (0 + 1)v = 1v = v = 0 + v$$

The first equality follows by VS8 and the last equality follows by VS?.

By the Cancellation Law, we have that $0v = 0$. Since v was arbitrary, we're done.

Exercise 1.2.12

Definitions and Setup:

- A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *even* if $f(t) = f(-t)$ for all $t \in \mathbb{R}$.
- The set of real-valued even functions is denoted E , where

$$E = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f(t) = f(-t), \forall t \in \mathbb{R}\}.$$

- We'll verify VS1-VS8 for E , with the operations of pointwise addition and scalar multiplication defined as:

$$(f + g)(x) = f(x) + g(x), \quad (\alpha f)(x) = \alpha \cdot f(x),$$

for all $f, g \in E$, $\alpha \in \mathbb{R}$, and $x \in \mathbb{R}$.

Verification of Vector Space Axioms:

1. **(VS1) Commutativity of Addition:** For $f, g \in E$, we need to show $f + g = g + f$. By definition of pointwise addition,

$$(f + g)(x) = f(x) + g(x) \quad \text{and} \quad (g + f)(x) = g(x) + f(x).$$

Since addition of real numbers is commutative ($f(x) + g(x) = g(x) + f(x)$), we have $(f + g)(x) = (g + f)(x)$. Thus, $f + g = g + f$.

2. **(VS2) Associativity of Addition:** For $f, g, h \in E$, we need to show $(f + g) + h = f + (g + h)$. By definition of pointwise addition,

$$((f + g) + h)(x) = (f + g)(x) + h(x) = (f(x) + g(x)) + h(x),$$

and

$$(f + (g + h))(x) = f(x) + (g + h)(x) = f(x) + (g(x) + h(x)).$$

Since addition of real numbers is associative ($(f(x) + g(x)) + h(x) = f(x) + (g(x) + h(x))$), we have $((f + g) + h)(x) = (f + (g + h))(x)$. Thus, $(f + g) + h = f + (g + h)$.

3. **(VS3) Existence of Additive Identity:** The zero function 0 , defined by $0(x) = 0$ for all $x \in \mathbb{R}$, acts as the additive identity. For $f \in E$, we need to show $f + 0 = f$. By definition of pointwise addition,

$$(f + 0)(x) = f(x) + 0(x) = f(x) + 0.$$

Since $f(x) + 0 = f(x)$ (additive identity of real numbers), we have $(f + 0)(x) = f(x)$. Thus, $f + 0 = f$.

4. **(VS4) Existence of Additive Inverses:** For $f \in E$, we need to show there exists $g \in E$ such that $f + g = 0$. Let $g = -f$, where $(-f)(x) = -f(x)$ for all $x \in \mathbb{R}$. By definition of pointwise addition,

$$(f + g)(x) = f(x) + g(x) = f(x) + (-f(x)) = 0.$$

Thus, $f + g = 0$, and $g \in E$ because $(-f)(-x) = -(f(-x)) = -f(x)$ (preserving evenness). Hence, the additive inverse of f exists and is $-f$.

5. **(VS5) Compatibility of Scalar Multiplication with the Scalar 1:** For $f \in E$, we need to show $1 \cdot f = f$. By definition of scalar multiplication,

$$(1 \cdot f)(x) = 1 \cdot f(x) = f(x).$$

Thus, $1 \cdot f = f$.

6. **(VS6) Compatibility of Scalar Multiplication with Scalar Multiplication of Scalars:** For $\alpha, \beta \in \mathbb{R}$ and $f \in E$, we need to show $(\alpha\beta) \cdot f = \alpha \cdot (\beta \cdot f)$. By definition of scalar multiplication,

$$((\alpha\beta) \cdot f)(x) = (\alpha\beta) \cdot f(x),$$

and

$$(\alpha \cdot (\beta \cdot f))(x) = \alpha \cdot ((\beta \cdot f)(x)) = \alpha \cdot (\beta \cdot f(x)).$$

Since multiplication of real numbers is associative ($(\alpha\beta) \cdot f(x) = \alpha \cdot (\beta \cdot f(x))$), we have $(\alpha\beta) \cdot f = \alpha \cdot (\beta \cdot f)$.

7. **(VS7) Distributivity of Scalar Multiplication with Respect to Vector Addition:** For $\alpha \in \mathbb{R}$ and $f, g \in E$, we need to show $\alpha \cdot (f + g) = \alpha \cdot f + \alpha \cdot g$. By definition of scalar multiplication and pointwise addition,

$$(\alpha \cdot (f + g))(x) = \alpha \cdot (f + g)(x) = \alpha \cdot (f(x) + g(x)),$$

and

$$(\alpha \cdot f + \alpha \cdot g)(x) = (\alpha \cdot f)(x) + (\alpha \cdot g)(x) = \alpha \cdot f(x) + \alpha \cdot g(x).$$

Since $\alpha \cdot (f(x) + g(x)) = \alpha \cdot f(x) + \alpha \cdot g(x)$ (distributive property of real numbers), we have $\alpha \cdot (f + g) = \alpha \cdot f + \alpha \cdot g$.

8. **(VS8) Distributivity of Scalar Multiplication with Respect to Scalar Addition:** For $\alpha, \beta \in \mathbb{R}$ and $f \in E$, we need to show $(\alpha + \beta) \cdot f = \alpha \cdot f + \beta \cdot f$. By definition of scalar multiplication,

$$((\alpha + \beta) \cdot f)(x) = (\alpha + \beta) \cdot f(x),$$

and

$$(\alpha \cdot f + \beta \cdot f)(x) = (\alpha \cdot f)(x) + (\beta \cdot f)(x) = \alpha \cdot f(x) + \beta \cdot f(x).$$

Since $(\alpha + \beta) \cdot f(x) = \alpha \cdot f(x) + \beta \cdot f(x)$ (distributive property of real numbers), we have $(\alpha + \beta) \cdot f = \alpha \cdot f + \beta \cdot f$.

Conclusion: Since all vector space axioms (VS1 through VS8) are satisfied, the set of real-valued even functions forms a vector space over $\mathbb{R}$ under pointwise addition and scalar multiplication.

Notice that the set of all even function is NOT a field since for example the (multiplicative) inverse of $f(x) = x^2$, namely, $g(x) = \frac{1}{x^2}$, is undefined at 0. Thus, we don't have a multiplicative inverse.

In particular, under pointwise multiplication, no function with a zero is going to have a multiplicative inverse.

1.1 Definition after Exercise 1.3.22

Remember how the union of two subspaces was no longer a subspace. This is an attempt to fix it. To show that $S_1 + S_2$ is a subspace, we need to check if it satisfies the three conditions for a subspace of a vector space.

Notice that $0 \in S_1$ and $0 \in S_2$ because they are subspaces. Thus, $0 + 0 = 0 \in S_1 + S_2$.

Let's now show closure under addition.

Let $u_1 + v_1 \in S_1 + S_2$ and $u_2 + v_2 \in S_1 + S_2$, where $u_1, u_2 \in S_1$ and $v_1, v_2 \in S_2$. We need to show that the sum of these vectors is also in $S_1 + S_2$.

$$(u_1 + v_1) + (u_2 + v_2) = (u_1 + u_2) + (v_1 + v_2).$$

Since S_1 is closed under addition (as it is a subspace), $u_1 + u_2 \in S_1$. Similarly, since S_2 is closed under addition, $v_1 + v_2 \in S_2$.

Thus:

$$(u_1 + u_2) + (v_1 + v_2) \in S_1 + S_2.$$

Therefore, $S_1 + S_2$ is closed under addition.

Finally, let's show closure under scalar multiplication.

Let $u + v \in S_1 + S_2$, where $u \in S_1$ and $v \in S_2$, and let $c \in \mathbb{F}$ be any scalar. We need to show that $c \cdot (u + v) \in S_1 + S_2$.

$$c \cdot (u + v) = (c \cdot u) + (c \cdot v).$$

Since S_1 is closed under scalar multiplication, $c \cdot u \in S_1$. Similarly, since S_2 is closed under scalar multiplication, $c \cdot v \in S_2$.

Thus:

$$(c \cdot u) + (c \cdot v) \in S_1 + S_2.$$

Therefore, $S_1 + S_2$ is closed under scalar multiplication.

Since $S_1 + S_2$ is non-empty, closed under addition, and closed under scalar multiplication, we conclude that $S_1 + S_2$ is indeed a subspace of V .

1.2 Exercise 1.3.10

We are given two sets:

$$W_1 = \{(a_1, a_2, \dots, a_n) \in \mathbb{F}^n : a_1 + a_2 + \dots + a_n = 0\}$$

and

$$W_2 = \{(a_1, a_2, \dots, a_n) \in \mathbb{F}^n : a_1 + a_2 + \dots + a_n = 1\}.$$

To prove that W_1 is a subspace of $\mathbb{F}^n$, we need to verify that it contains the zero vector, is closed under addition, and is closed under scalar multiplication.

We begin by checking if the zero vector is in W_1 . The zero vector in $\mathbb{F}^n$ is $(0, 0, \dots, 0)$. Substituting into the defining equation of W_1 , we have:

$$0 + 0 + \dots + 0 = 0.$$

Thus, the zero vector is in W_1 .

Next, we verify if W_1 is closed under addition. Let $\mathbf{u} = (u_1, u_2, \dots, u_n) \in W_1$ and $\mathbf{v} = (v_1, v_2, \dots, v_n) \in W_1$. Then, by the definition of W_1 ,

$$u_1 + u_2 + \dots + u_n = 0 \quad \text{and} \quad v_1 + v_2 + \dots + v_n = 0.$$

Consider their sum $\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$. The sum of the components is:

$$(u_1 + v_1) + (u_2 + v_2) + \dots + (u_n + v_n) = (u_1 + u_2 + \dots + u_n) + (v_1 + v_2 + \dots + v_n) = 0 + 0 = 0.$$

Thus, $\mathbf{u} + \mathbf{v} \in W_1$, proving that W_1 is closed under addition.

Finally, we verify if W_1 is closed under scalar multiplication. Let $\mathbf{u} = (u_1, u_2, \dots, u_n) \in W_1$ and $c \in \mathbb{F}$. By the definition of W_1 ,

$$u_1 + u_2 + \dots + u_n = 0.$$

Consider the scalar multiple $c\mathbf{u} = (cu_1, cu_2, \dots, cu_n)$. The sum of the components is:

$$cu_1 + cu_2 + \dots + cu_n = c(u_1 + u_2 + \dots + u_n) = c \cdot 0 = 0.$$

Thus, $c\mathbf{u} \in W_1$, proving that W_1 is closed under scalar multiplication.

Since W_1 contains the zero vector, is closed under addition, and is closed under scalar multiplication, we conclude that W_1 is a subspace of $\mathbb{F}^n$.

Now, let us show that W_2 is not a subspace of $\mathbb{F}^n$. To do this, we check if it contains the zero vector. The zero vector in $\mathbb{F}^n$ is $(0, 0, \dots, 0)$. Substituting into the defining equation of W_2 , we get:

$$0 + 0 + \dots + 0 = 0.$$

However, this does not satisfy the equation defining W_2 , which requires the sum to be 1. Therefore, the zero vector is not in W_2 .

Since W_2 does not contain the zero vector, it cannot be a subspace of $\mathbb{F}^n$.

Thus, we conclude that W_1 is a subspace, but W_2 is not.

2 1.5.13

Lemma 2.1. Let V be a vector space over a field of characteristic not equal to two.

- (a) Let u and v be distinct vectors in V . Then $\{u, v\}$ is linearly independent if and only if $\{u+v, u-v\}$ is linearly independent.
- (b) Let u, v , and w be distinct vectors in V . Then $\{u, v, w\}$ is linearly independent if and only if $\{u+v, u+w, v+w\}$ is linearly independent.

Proof. (a) We need to show that $\{u, v\}$ is linearly independent if and only if $\{u+v, u-v\}$ is linearly independent.

Forward direction: Assume $\{u, v\}$ is linearly independent. Consider a linear combination

$$a(u+v) + b(u-v) = 0.$$

Expanding, we get

$$au + av + bu - bv = 0 \implies (a+b)u + (a-b)v = 0.$$

Since $\{u, v\}$ is linearly independent, we must have $a+b=0$ and $a-b=0$. Solving these, we get $a=b=0$, proving that $\{u+v, u-v\}$ is linearly independent.

Reverse direction: Assume $\{u+v, u-v\}$ is linearly independent. Consider a linear combination

$$au + bv = 0.$$

Rewrite this as

$$a\left(\frac{u+v}{2} + \frac{u-v}{2}\right) + b\left(\frac{u+v}{2} - \frac{u-v}{2}\right) = 0,$$

which implies that the only solution is $a=b=0$. Thus, $\{u, v\}$ is linearly independent.

(b) We need to show that $\{u, v, w\}$ is linearly independent if and only if $\{u+v, u+w, v+w\}$ is linearly independent.

Forward direction: Assume $\{u, v, w\}$ is linearly independent. Consider a linear combination

$$a(u+v) + b(u+w) + c(v+w) = 0.$$

Expanding, we get

$$au + av + bu + bw + cv + cw = 0 \implies (a+b)u + (a+c)v + (b+c)w = 0.$$

Since $\{u, v, w\}$ is linearly independent, we must have $a+b=0$, $a+c=0$, and $b+c=0$. Solving these equations, we get $a=b=c=0$, proving that $\{u+v, u+w, v+w\}$ is linearly independent.

Reverse direction: Assume $\{u+v, u+w, v+w\}$ is linearly independent. Consider a linear combination

$$au + bv + cw = 0.$$

Expressing each vector in terms of sums:

$$a\left(\frac{(u+v) + (u-v)}{2}\right) + b\left(\frac{(u+w) + (w-u)}{2}\right) + c\left(\frac{(v+w) + (w-v)}{2}\right) = 0,$$

which yields $a=b=c=0$. Thus, $\{u, v, w\}$ is linearly independent. □